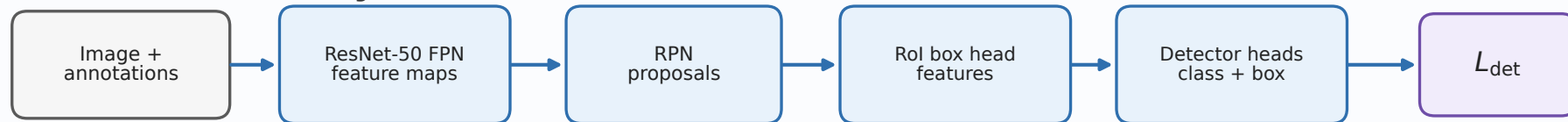


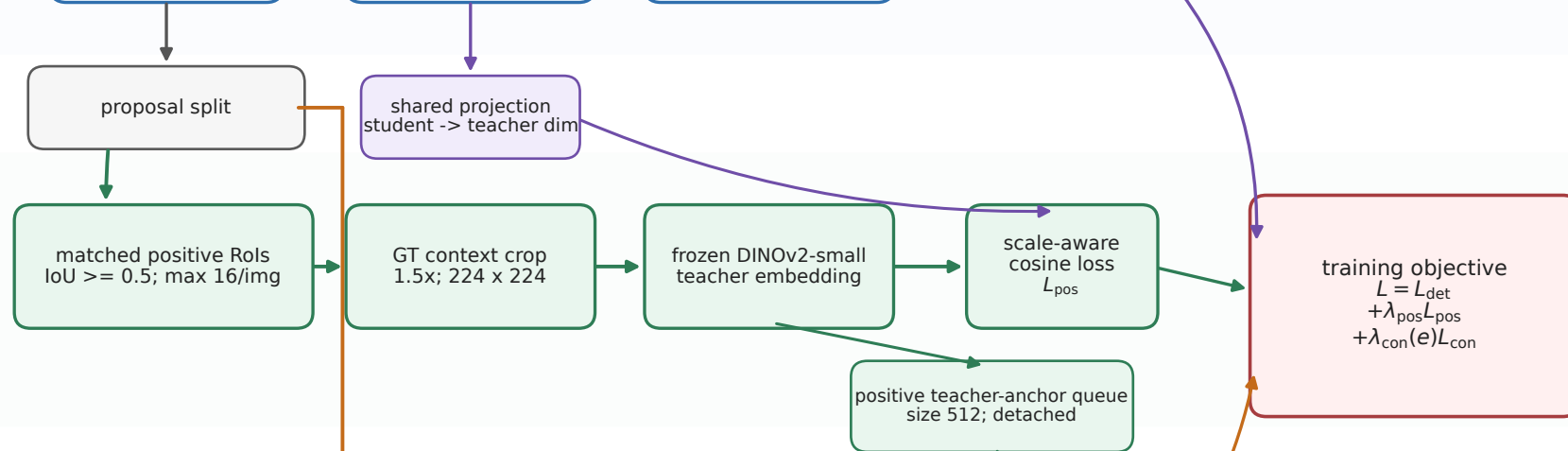
HN-SARD training pipeline

student keeps the detector path; teacher and contrastive queue are training-only regularizers

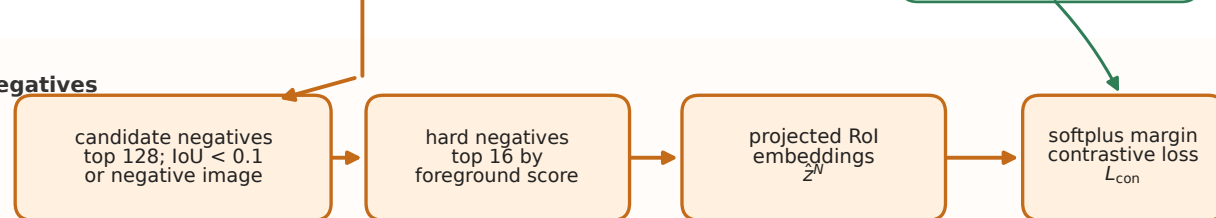
Student detector branch: full-image Faster R-CNN



Positive region distillation: frozen teacher crops



Online hard-negative contrast: foreground-like negatives



selected setting: $\lambda_{\text{pos}} = 0.0025$, $\lambda_{\text{con}} = 0.02$ after 5-epoch warmup

inference uses only the trained detector